<u>Project Logbook — Personalized Weather Station</u>

Phase 1 — Project Ideation & Planning (Late January – Early February)

- Project initiated in late January 2026
- Initial goal was to create a manufacturable embedded environmental monitoring system capable of measuring temperature, humidity, and atmospheric pressure
- Early concepts explored both Raspberry Pi and standalone embedded approaches before committing to a custom PCB architecture
- Selected the BME280 as the primary environmental sensor due to its ability to measure temperature, humidity, and barometric pressure in a single package
- Established GitHub repository for version control, documentation, and organization
- Created the initial project definition document outlining objectives, constraints, and proposed system functionality

Phase 2 — Proof of Concept & Sensor Validation (February – Early March)

- Used a Raspberry Pi temporarily as a proof-of-concept platform to validate environmental sensing and data collection
- Initial BME280 units were damaged during testing due to incorrect pin labeling and wiring confusion
- Multiple communication and wiring configurations were tested while troubleshooting sensor reliability
- DHT22 was temporarily used to validate the sensing workflow while replacement BME280 units were obtained
- Basic GUI and sensor visualization concepts were tested during this stage to validate real-time environmental monitoring

● Proof-of-concept phase confirmed the feasibility of the environmental monitoring system before transitioning to standalone PCB development

Phase 3 — Transition to Custom PCB Architecture (March 5th)

● During design review it became clear that a Raspberry Pi and touchscreen-based system would not reflect a realistic manufacturable embedded product

● Team agreed that a standalone embedded PCB design would better align with course goals and real-world engineering practices

● Decision made to redesign the system around the ATmega328P-A microcontroller

● Replaced the touchscreen GUI concept with USB serial communication and future phone notification capability

● Replacement BME280 sensor was correctly wired and validated, officially replacing the DHT22 proof-of-concept setup

● Project scope shifted toward a manufacturable PCB-focused embedded system

Phase 4 — Schematic Design & Embedded Hardware Development (March – April)

● Schematic designed in KiCad using the ATmega328P-A as the primary processing unit

● Integrated the BME280 environmental sensor for temperature, humidity, and atmospheric pressure monitoring

● Used the available KiCad BME280 footprint and configured the sensor for I2C communication by tying the CSB pin high

● External crystal oscillator and 22pF load capacitors added to improve MCU timing stability and communication reliability

● Selected passive component values directly from component datasheets for stable operation

● Added decoupling capacitors to VCC, AVCC, and sensor power lines to reduce electrical noise

● Added ISP programming header (J2) for direct firmware flashing to the ATmega328P-A

● Added test points for probing, debugging, and future expansion

● Added status LED with current-limiting resistor for visual debugging and activity indication

Phase 5 — PCB Layout & Design Review (April 1st)

● Initial PCB layout completed and reviewed collaboratively

● Design review identified missing decoupling and bulk capacitors which were later added to improve stability

● Routing constraints on a standard 2-layer PCB created congestion and limited trace flexibility

● Experimented with converting the design to a 4-layer stackup to learn inner-layer routing and via assignment techniques

● Learned how to assign vias to specific layers and route signals across internal copper layers

● ERC and DRC checks were performed and all violations were resolved

● Generated 3D board renders to validate physical layout, spacing, and component placement

Phase 6 — Manufacturing Preparation (Late April – Early May)

● Generated Gerber fabrication files for all PCB layers

● Exported drill files for PCB manufacturing

● Compiled BOM containing component values, quantities, and part numbers

● Prepared PCB ordering instructions for fabrication services such as JLCPCB and PCBWay

● Organized repository structure with hardware files, fabrication outputs, documentation, and software resources

● Verified the project was manufacturing-ready and capable of future assembly

Phase 7 — Final System Integration & Presentation Preparation (May 7th)

● Push button added to support user-triggered notification functionality

● USB serial communication workflow finalized between the ATmega328P-A and external computer application

● Python-based data visualization and future weather trend prediction concepts integrated into the final presentation workflow

● Discussed future implementation ideas involving ntfy notifications, wireless monitoring, and predictive weather analysis

● Final presentation prepared demonstrating the complete engineering workflow from concept to manufacturable PCB

● Project finalized and submitted on May 7th, 2026

Lessons Learned

● Hardware reliability must always be verified before full system integration

● Incorrect sensor pin labeling can quickly damage components and delay development

● Early proof-of-concept platforms are useful for validating ideas before committing to final hardware architecture

● Design reviews are extremely valuable and can identify missing hardware before manufacturing

● Proper decoupling and power stability are critical in embedded system design

- Learning 4-layer PCB routing significantly improved understanding of professional PCB design workflows
- Real engineering development is iterative and improves through debugging, redesign, and continuous refinement
- This project provided practical experience with schematic capture, PCB routing, manufacturing preparation, debugging, and embedded system integration